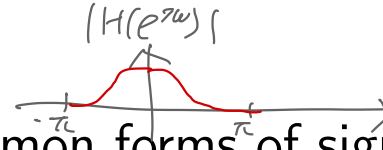


$$y[n] = h[n] * x[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

Filter design:

- **Filtering** *frequency selective filtering* is one of the most common forms of signal processing.



$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

- A filter can be equivalently described by $h[n]$, $H(z)$, $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$ in magnitude & phase, or $H(e^{j\omega}) = A(\omega)e^{j\phi(\omega)}$ for “signed-magnitude” $A(\omega) \in \mathbb{R}$ and “continuous-phase” $\phi(\omega) \in \mathbb{R}$.
- When designing a filter $H(e^{j\omega})$, we are concerned with
 1. filtering performance:
 - want magnitude response $|H(e^{j\omega})|$ close to target,
 - want phase response $\angle H(e^{j\omega})$ to be “agreeable.”
 2. implementational issues:
 - complexity,
 - memory,
 - robustness to quantization effects.
- There are two main classes of digital filter
 1. *recursive* or “*IIR*” (infinite impulse response), and
 2. *non-recursive* or “*FIR*” (finite impulse response).

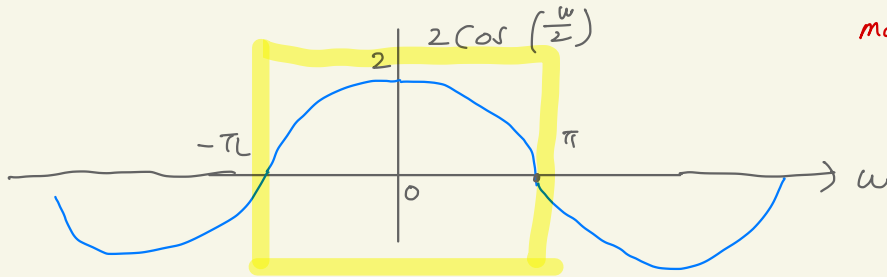
We will see advantages and disadvantages to each.

$$h[n] = f[n] + f[n-1]$$

$$H(e^{j\omega}) = 1 + e^{-j\omega} = e^{-j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}})$$

$$\frac{0 - \omega}{2} = -\frac{\omega}{2}$$

$$= \underbrace{2 \cos\left(\frac{\omega}{2}\right)}_{\text{magnitude}} \underbrace{e^{-j\frac{\omega}{2}}}_{\text{phase}}$$



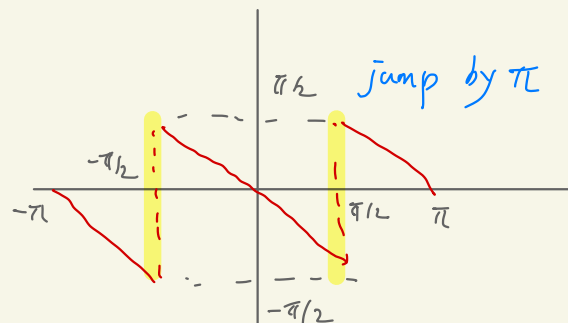
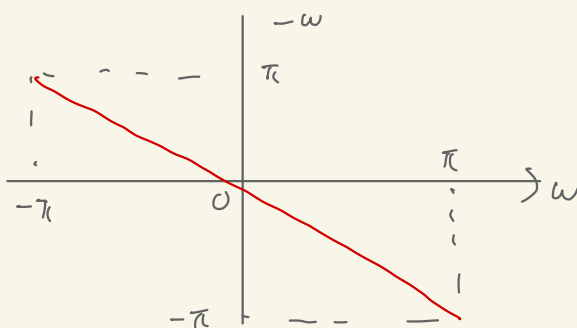
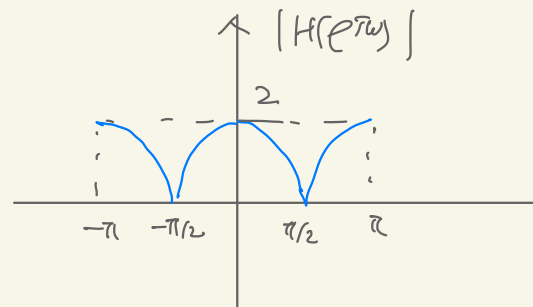
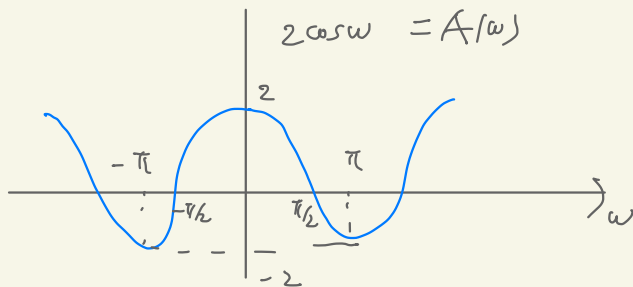
$$h[n] = f[n] + f[n-2]$$

$$H(e^{j\omega}) = 1 + e^{-j2\omega} = e^{-j\omega} (e^{j\omega} + e^{-j\omega})$$

$$\left(\frac{0 - 2\omega}{2} = -\omega\right) = 2 \cos \omega e^{-j\omega}$$

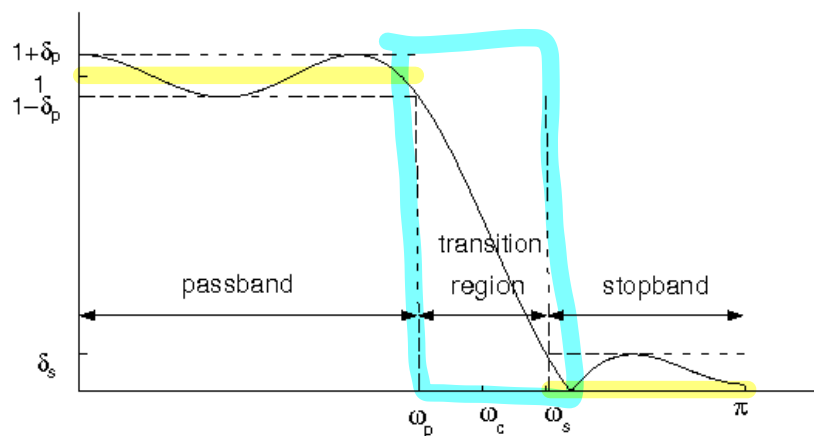
$$|2 \cos \omega| = |H(e^{j\omega})| = \begin{cases} 2 \cos \omega & |\omega| \leq \frac{\pi}{2} \\ -2 \cos \omega & \frac{\pi}{2} < |\omega| \leq \pi \end{cases}$$

"signed" magnitude



Desired magnitude response:

- The desired magnitude response $|D(e^{j\omega})|$ is often a LPF, HPF, BPF, or notch filter, though sometimes it is more complicated, e.g., a differentiator, raised-cosine, or Hilbert transformer.
- The error between $|D(e^{j\omega})|$ and the actual response $|H(e^{j\omega})|$ can be measured in different ways:
 - *peak deviations* $\{\delta_p, \delta_s\}$ in passband and stopband:



or $A_p \triangleq -20 \log_{10}(1 - \delta_p)$, $A_s \triangleq -20 \log_{10} \delta_s$ in dB.

- *squared error*, integrated over ω and possibly weighted:

$$\mathcal{E} = \int_{-\pi}^{\pi} W(\omega) (|H(e^{j\omega})| - |D(e^{j\omega})|)^2 d\omega$$

Beware: different error criteria lead to different “optimal” filter designs.

Desired phase response:

- Sometimes, we want that $H(e^{j\omega})$ matches a desired response $D(e^{j\omega})$ in phase (as well as magnitude).
- Other times, we want $H(e^{j\omega})$ giving “minimum” delay.
- More often, we desire a “*linear phase*” response

$$\phi(\omega) = -\omega\ell \text{ for some } \ell \in \mathbb{Z},$$

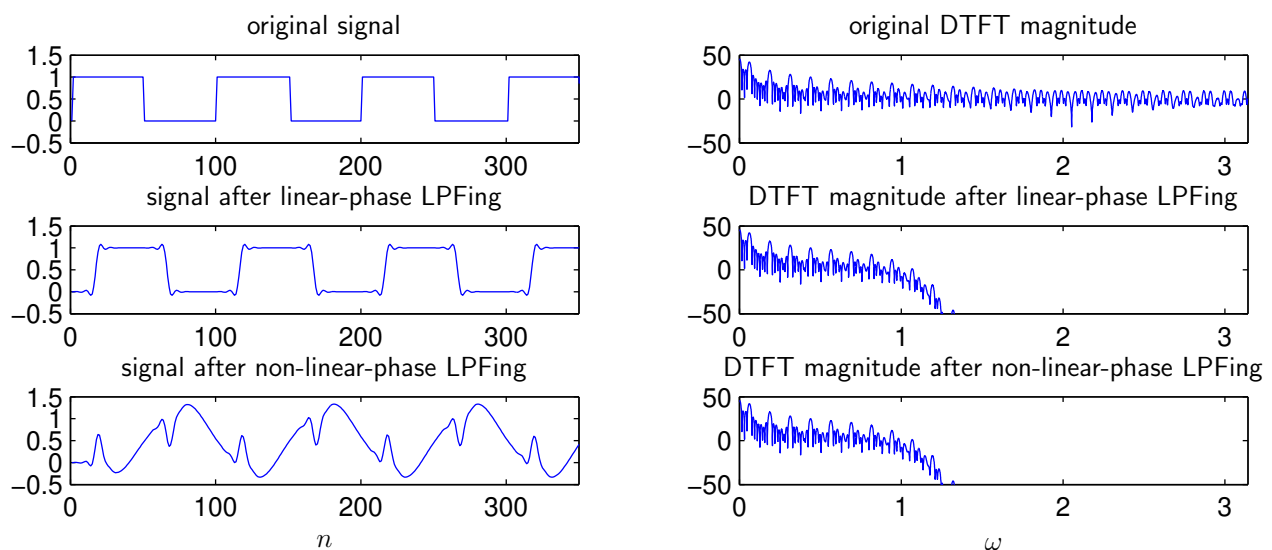
because the resulting filter

$$H(e^{j\omega}) = A(\omega)e^{j\phi(\omega)} = A(\omega)e^{-j\omega\ell}$$

alters the signed-magnitude response by $A(\omega)$ and delays the resulting signal by ℓ samples.

Trivial phase modification $\Rightarrow$ no phase distortion!

- Phase distortion can have a huge effect on the time-domain characteristics of the filtered signal:



and such distortion is unwanted in many applications.

Non-recursive filters:

$$x[n-k] = f[n-k] * x[n]$$

- Non-recursive digital filters typically have the MA form:

$$H(z) = \sum_{k=0}^N b_k z^{-k} \quad \text{where} \quad \begin{cases} N = \text{"order"} \\ N+1 = \text{"length"} \end{cases}$$

which results in the difference equation

$$\left(\sum_{k=0}^N b_k f[n-k] \right) * x[n] = y[n] = \sum_{k=0}^N b_k x[n-k]$$

moving average system

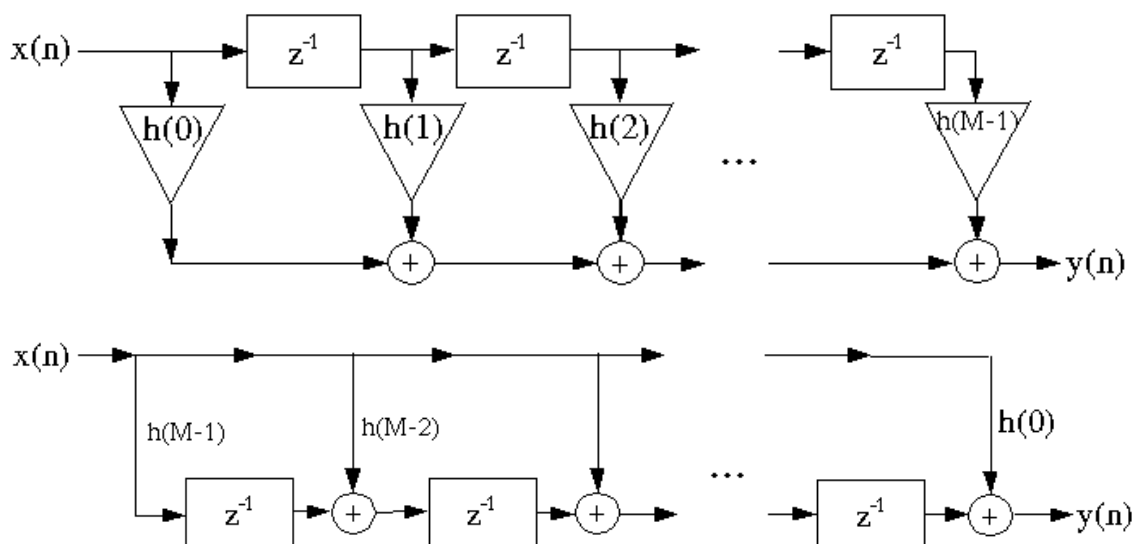
and thus the (causal) impulse response

$$h[k] = \begin{cases} b_k & \text{for } k = 0 \dots N \\ 0 & \text{else} \end{cases}$$

- Both memory and complexity grow with N . Since these must be finite, the impulse response length must be too:

$\Rightarrow$ "finite impulse response" (FIR) filter

- Implementation options:



$$H(z) = \frac{B(z)}{A(z)}$$

$$H(z)X(z) = \frac{1}{A(z)} B(z) X(z) = B(z) \frac{1}{A(z)} X(z)$$

z^{-1} delay unit

Recursive filters:

- Recursive digital filters typically have the ARMA form:

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 + \sum_{m=1}^M a_m z^{-m}}$$

with "order" = $\max(M, N)$

which results in the difference equation

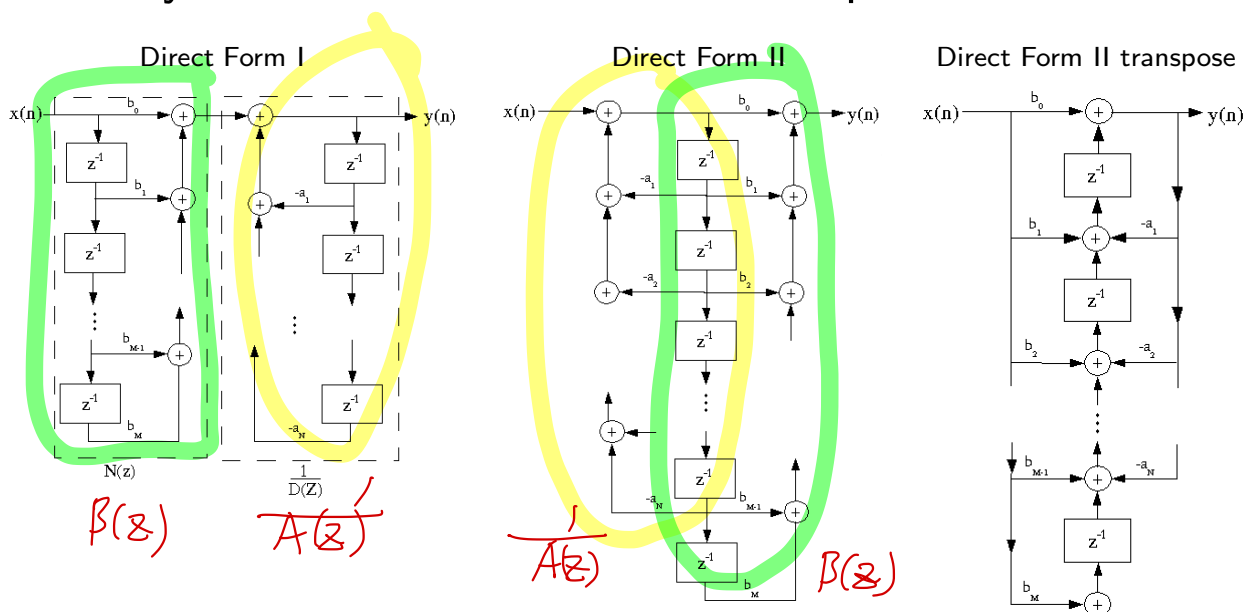
$$y[n] = \underbrace{\sum_{k=0}^N b_k x[n-k]}_{\text{MA}} - \underbrace{\sum_{m=1}^M a_m y[n-m]}_{\text{AR}}$$

ARMA system
auto regressive & moving average

- The impulse response $h[n]$ has a complicated expression and infinite length.

$\Rightarrow$ "infinite impulse response" (IIR) filter

- There are several ways to implement this. While identical in theory, some are more sensitive to quantization.



Note that the minimum number of delays is $\max(M, N)$, i.e., the filter order.

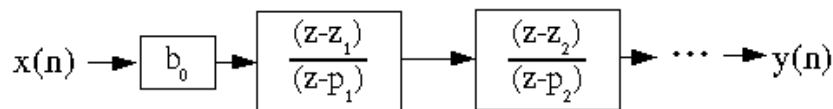
Recursive filters (cont.):

- Another way to write ARMA $H(z)$ is to factor the numerator and denominator as

$$H(z) = \frac{b_0 \prod_{k=1}^N (1 - z_k z^{-1})}{\prod_{m=1}^M (1 - p_m z^{-1})}$$

with “zeros” $\{z_k\}_{k=1}^N$ and “poles” $\{p_m\}_{m=1}^M$.

- This factorization suggests the cascade implementation

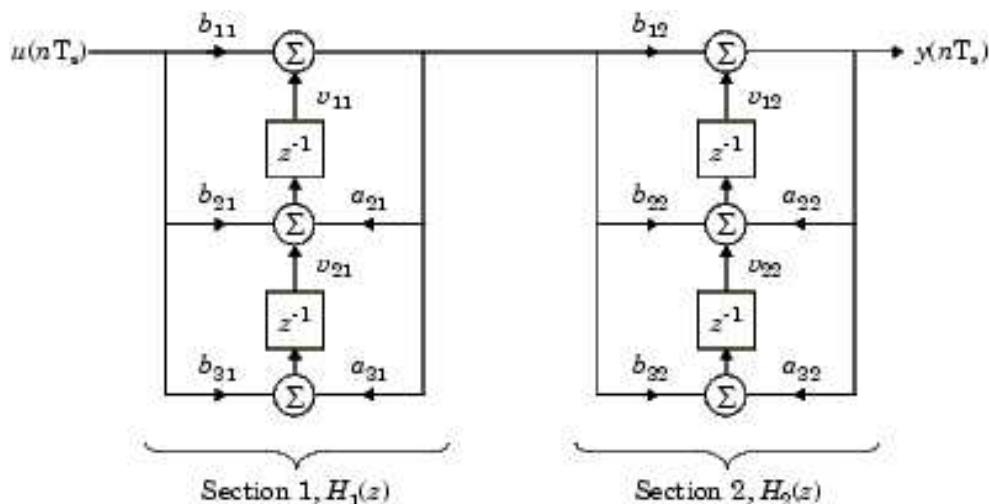


which is very robust to quantization effects.

- When $h[n]$ is real-valued, we can leverage the fact that both $\{z_k\}_{k=1}^N$ and $\{p_m\}_{m=1}^M$ come in complex conjugate pairs, and cascade real-coefficient 2nd-order sections

$$b_0 \frac{(z - z_n)(z - z_n^*)}{(z - p_n)(z - p_n^*)} = b_0 \frac{z^2 - 2 \operatorname{Re}\{z_n\}z + |z_n|^2}{z^2 - 2 \operatorname{Re}\{p_n\}z + |p_n|^2}$$

as follows:



$$H(z) = \frac{1}{(p_1 z^{-1} - 1)(p_1^* z^{-1} - 1)(p_2 z^{-1} - 1)(p_2^* z^{-1} - 1)(p_3 z^{-1} - 1)(p_3^* z^{-1} - 1)}$$

poles $p_1, p_1^*, p_2, p_2^*, p_3, p_3^*$

$$H_k(z) = \frac{1}{(p_k z^{-1} - 1)(p_k^* z^{-1} - 1)} \quad k=1, 2, 3$$

$$H(z) = H_1(z) H_2(z) H_3(z)$$

$$H(z) = \frac{1}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_6 z^{-6}}$$

$$\tilde{H}(z) = \frac{1}{\tilde{a}_0 + \tilde{a}_1 z^{-1} + \tilde{a}_2 z^{-2} + \dots + \tilde{a}_6 z^{-6}}$$

$$H_1(z) = \frac{1}{c_0 + c_1 z^{-1} + c_2 z^{-2}}$$

$$\tilde{H}_1(z) = \frac{1}{\tilde{c}_0 + \tilde{c}_1 z^{-1} + \tilde{c}_2 z^{-2}}$$

Advantages of FIR and IIR filters:

The advantages of *IIR filters* are:

- much less computation/memory needed to meet target magnitude response,
- better for approximating analog filters,

and the advantages of *FIR filters* are:

- conceptually simpler,
- can have linear phase,
- always stable,
- relatively insensitive to quantization noise,
- easy to make “adaptive,”
- can implement using fast convolution.

In the end, the “best” choice depends on the application!